# DNS (Domain Name System)

## What is DNS?

DNS (Domain Name System) translates domain names → IP addresses.

Example:
example.com → 93.184.215.14

Reason:

- Humans remember domain names
- Computers communicate using IP addresses

So DNS acts like the Internet's phonebook.

DNS operates at:
Application Layer (Layer 7)

## DNS Records

### 1. A Record

A (Address) Record

Maps:
Hostname → IPv4 Address

Example:
example.com → 172.17.2.172

### 2. AAAA Record

AAAA (Quad-A) Record

Maps:
Hostname → IPv6 Address

Example:
example.com → 2606:2800:21f:cb07:6820:80da:af6b:8b2c

### 3. CNAME Record

CNAME (Canonical Name)

Maps:
Domain → Another Domain

Example:
www.example.com → example.com

Used for domain aliases.

**4. MX Record**
MX (Mail Exchange)
Specifies the mail server responsible for emails for a domain.

Example:
example.com → mail.example.com

Used when sending emails like:
user@example.com

Mail server queries MX records to find where to send the mail.

## DNS Query Example

When a user types:
www.example.com

Process:
1. Browser sends DNS query
2. DNS server checks A record
3. Returns IP address
4. Browser connects to the server

Example result:
www.example.com → 93.184.215.14

## DNS Lookup Using Command Line

Tool: nslookup

Example:

```
nslookup www.example.com
```

Output:

```
user@TryHackMe$ nslookup www.example.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   www.example.com
Address: 93.184.215.14
Name:   www.example.com
Address: 2606:2800:21f:cb07:6820:80da:af6b:8b2c
```

# Domain Registration & WHOIS

## Who Controls DNS Records

The person or organization that registers a domain name has the authority to configure its DNS records.

Example:

- If someone registers:
  - example.com
- They can configure records like:
  - A record
  - AAAA record
  - MX record
  - CNAME
  - other DNS records

These records determine how the domain behaves on the Internet.

## Domain Registration

To use a domain name:

1. Check if the domain is available
2. Register it through a domain registrar
3. Pay a yearly fee
4. Provide contact information

Registration duration: Usually 1+ years

If you don't renew, the domain may become available again.

## Domain Registrar

A registrar is a company that sells domain names.

functions:

- Register domains
- Manage DNS records
- Handle renewals

Example registrar:
GoDaddy

## WHOIS

WHOIS is a protocol/database that provides information about domain ownership.

Example command:

```
whois example.com
```

Example WHOIS Output

```
Registrar: GoDaddy.com, LLC
Creation Date: 1993-04-02
Updated Date: 2017-07-05
Expiration Date: 2026-10-20
Registrant Name: Registration Private
```

```
user@TryHackMe$ whois [REDACTED].com
[...]
Domain Name: [REDACTED].COM
Registry Domain ID: [REDACTED]
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2017-07-05T16:02:43Z
Creation Date: 1993-04-02T00:00:00Z
Registrar Registration Expiration Date: 2026-10-20T14:56:17Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
[...]
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
[...]
```

# HTTP(S): Accessing the Web

## What is HTTP?

HTTP (Hypertext Transfer Protocol) is a protocol used for communication between:

Web Browser (Client) ↔ Web Server

It defines how web pages and resources are requested and delivered.

Example resources:

- HTML pages
- Images
- CSS files
- JavaScript files

HTTPS = HTTP + Security

The S means the connection is encrypted using TLS/SSL.

## HTTP Methods

HTTP uses methods (verbs) to tell the server what action to perform.

| | |
|---|---|
| GET | Retrieve data from server |
| POST | Submit data to server |
| PUT | Create or overwrite a resource |
| DELETE | Remove a resource |

**1. GET Example**
Retrieve homepage:

```
GET / HTTP/1.1
```

Retrieve specific file:

```
GET /file.html HTTP/1.1
```

**2. POST Example**
Used when submitting data:

- login forms
- uploading files
- submitting forms

Example:

```
POST /login HTTP/1.1
```

**3. PUT Example**
Creates or replaces a resource on the server.

Example:

```
PUT /file.txt HTTP/1.1
```

**4. DELETE Example**
Removes a resource from the server.

Example:

```
DELETE /file.txt HTTP/1.1
```

**Example HTTP Request (Manual)**
You can manually send HTTP requests using telnet.

Connect to server:

```
telnet 10.48.144.138 80
```

Send request:

```
GET /file.http HTTP/1.1
Host: anything
```

Press Enter twice to finish the request.

# FTP (File Transfer Protocol)

FTP (File Transfer Protocol) is a protocol used to transfer files between a client and a server.

Used for:

- uploading files
- downloading files
- managing files on a server

FTP is often faster for file transfer than HTTP when conditions are equal.

## Common FTP Commands

| | |
|---|---|
| USER | Send username |
| PASS | Send password |
| RETR | Download a file |
| STOR | Upload a file |
| LIST | List files on server |

Example meaning:

RETR file.txt → download file
STOR file.txt → upload file

## Example FTP Connection

Connect to server:

```
ftp 10.48.144.138
```

Anonymous Login:
Some FTP servers allow anonymous login.

Example:

```
Username: anonymous
Password: (blank)
```

Used for public file sharing.

## Example FTP Session

### Step 1 – Connect to server

```
ftp 10.48.144.138
```

### Step 2 – Login

```
USER anonymous
PASS (empty)
```

### Step 3 – List files

```
ls
```

Example output:
coffee.txt
flag.txt
Tea.txt

### Step 4 – Switch file mode

```
type ascii
```

ASCII mode is used for:

- Text files

Binary mode is used for:

- Images
- Executables
- Archives

### Step 5 – Download file

```
get coffee.txt
```

Downloads coffee.txt from server

```
user@TryHackMe$ ftp 10.49.172.97
Connected to 10.49.172.97 (10.49.172.97).
220 (vsFTPd 3.0.5)
Name (10.49.172.97:strategos): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
227 Entering Passive Mode (10,10,41,192,134,10).
150 Here comes the directory listing.
-rw-r--r--    1 0        0            1480 Jun 27 08:03 coffee.txt
-rw-r--r--    1 0        0              14 Jun 27 08:04 flag.txt
-rw-r--r--    1 0        0            1595 Jun 27 08:05 tea.txt
226 Directory send OK.
ftp> type ascii
200 Switching to ASCII mode.
ftp> get coffee.txt
local: coffee.txt remote: coffee.txt
227 Entering Passive Mode (10,10,41,192,57,100).
150 Opening BINARY mode data connection for coffee.txt (1480 bytes).
WARNING! 47 bare linefeeds received in ASCII mode
File may not have transferred correctly.
226 Transfer complete.
1480 bytes received in 8e-05 secs (18500.00 Kbytes/sec)
ftp> quit
221 Goodbye.
```

# SMTP (Simple Mail Transfer Protocol)

## What is SMTP?

SMTP (Simple Mail Transfer Protocol) is used to send emails between mail clients and mail servers or between mail servers.

**Communication flow:**

Mail Client → SMTP Server → Recipient Mail Server

**Example usage:**

- Sending email from Outlook / Gmail client
- Mail server transferring email to another server

**Default Port:** 25

## How SMTP Works

**Basic process of sending an email:**

1. Client connects to SMTP server
2. Client introduces itself
3. Sender address is specified
4. Recipient address is specified
5. Email content is sent
6. Server confirms delivery

**Important SMTP Commands**

| | |
|---|---|
| HELO / EHLO | Start SMTP session |
| MAIL FROM | Specify sender email |
| RCPT TO | Specify recipient email |
| DATA | Begin email message content |
| . | End of email message |
| QUIT | Close SMTP session |

## Example SMTP Session (Telnet)

**1. Connect to SMTP server:**

```
telnet 10.49.172.97 25
```

Server greeting:
220 example.thm ESMTP Exim

## 2. Start communication:

```
HELO client.thm
```

Server response:
250 Hello client.thm

## 3. Sending an Email

### Step 1 — Sender address

```
MAIL FROM: <user@client.thm>
```

Server response:
250 OK

### Step 2 — Recipient address

```
RCPT TO: <strategos@server.thm>
```

Server response:
250 Accepted

### Step 3 — Start email content

```
DATA
```

Server response:
354 Enter message

### Step 4 — Write email

Example message:

```
From: user@client.thm
To: strategos@server.thm
Subject: Telnet email

Hello. I am using telnet to send you an email!
```

### Step 5 — End message

SMTP uses:

```
.
```

A single dot on a new line to end the message.

```
user@TryHackMe$ telnet 10.49.172.97 25
Trying 10.49.172.97...
Connected to 10.49.172.97.
Escape character is '^]'.
220 example.thm ESMTP Exim 4.95 Ubuntu Thu, 27 Jun 2024 16:18:09 +0000
HELO client.thm
250 example.thm Hello client.thm [10.11.81.126]
MAIL FROM: <user@client.thm>
250 OK
RCPT TO: <strategos@server.thm>
250 Accepted
DATA
354 Enter message, ending with "." on a line by itself
From: user@client.thm
To: strategos@server.thm
Subject: Telnet email

Hello. I am using telnet to send you an email!
.
250 OK id=1sMrpq-0001Ah-UT
QUIT
221 example.thm closing connection
Connection closed by foreign host.
```

Server response:
250 OK

**Step 6 — Close connection**

```
QUIT
```

**SMTP real use case:**
Real email systems use SMTP for:

- sending email
- server-to-server delivery

**But email retrieval uses different protocols:**

- POP3 Download emails
- IMAP Synchronize emails

# POP3 (Post Office Protocol v3)

## Email Workflow

**Email communication usually uses two protocols:**
SMTP: Sends email to mail servers
POP3: Retrieves email from mail server

**Default port TCP:** 110

**Command to connect:**

```
telnet <IP> 110
```

Example:

```
telnet 10.49.172.97 110
```

## POP3 Authentication Flow

Typical login sequence:

```
USER <username>
PASS <password>
```

Example:

```
USER strategos
PASS password123
```

Server response:
+OK Logged in

## POP3 Commands

| | |
|---|---|
| USER username | Identify user |
| PASS password | Authenticate |
| STAT | Show number of messages + total size |
| LIST | List all emails with sizes |
| RETR n | Retrieve message number |
| DELE n | Delete message |
| QUIT | End session |

## Example POP3 Session

**1. Connect:**

```
telnet 10.49.172.97 110
```

**2. Login:**

```
USER strategos
PASS password
```

**3. Check mailbox:**

```
STAT
```

Example output:
+OK 3 1264

Meaning:
3 emails
Total size = 1264 bytes

**4. List emails:**

```
LIST
```

```
user@TryHackMe$ telnet 10.49.172.97 110
Trying 10.49.172.97...
Connected to 10.49.172.97.
Escape character is '^]'.
+OK [XCLIENT] Dovecot (Ubuntu) ready.
AUTH
+OK
PLAIN
.
USER strategos
+OK
PASS
+OK Logged in.
STAT
+OK 3 1264
LIST
+OK 3 messages:
1 407
2 412
3 445
.
RETR 3
+OK 445 octets
Return-path: <user@client.thm>
Envelope-to: strategos@server.thm
Delivery-date: Thu, 27 Jun 2024 16:19:35 +0000
Received: from [10.11.81.126] (helo=client.thm)
        by example.thm with smtp (Exim 4.95)
        (envelope-from <user@client.thm>)
        id 1sMrpq-0001Ah-UT
        for strategos@server.thm;
        Thu, 27 Jun 2024 16:19:35 +0000
From: user@client.thm
To: strategos@server.thm
Subject: Telnet email

Hello. I am using telnet to send you an email!
.
QUIT
+OK Logging out.
Connection closed by foreign host.
```

Output:
1 407
2 412
3 445

**5. Retrieve an email:**

```
RETR 3
```

Server returns the full email content.

## IMAP (Internet Message Access Protocol)

IMAP solves the POP3 limitations:

- Downloads emails to one device
- Often deletes email from server
- Not suitable for multiple devices

IMAP solution

- Keeps emails stored on the server
- Allows synchronization across multiple devices

Example devices:

- Desktop PC
- Laptop
- Smartphone
- Tablet

All devices will show the same mailbox state.

**Port: 143**

Connection example:

```
telnet <IP> 143
```

Example:

```
telnet 10.49.172.97 143
```

## Example IMAP Session

### 1. Connect:

```
telnet 10.10.41.192 143
```

Server response:
* OK Dovecot ready

### 2. Login:

```
A LOGIN strategos password
```

Server:
A OK Logged in

### 3. Select mailbox:

```
B SELECT inbox
```

Server shows mailbox info:
* 4 EXISTS
* 0 RECENT
* OK [UNSEEN 2]

Meaning:

| | |
|---|---|
| EXISTS | Total messages |
| RECENT | New messages |
| UNSEEN | Unread messages |

### 4. Retrieve an Email

```
C FETCH 3 body[]
```

Retrieve email number 3
Return header + body

Example message:

```
From: user@client.thm
To: strategos@server.thm
Subject: Telnet email
```

```
user@TryHackMe$ telnet 10.10.41.192 143
Trying 10.10.41.192...
Connected to 10.10.41.192.
Escape character is '^]'.
* OK [CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE LITERAL+ STARTTLS
A LOGIN strategos
A OK [CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE SORT SORT=DISPLAY
B SELECT inbox
* FLAGS (\Answered \Flagged \Deleted \Seen \Draft)
* OK [PERMANENTFLAGS (\Answered \Flagged \Deleted \Seen \Draft \*)] Flags permitted.
* 4 EXISTS
* 0 RECENT
* OK [UNSEEN 2] First unseen.
* OK [UIDVALIDITY 1719824692] UIDs valid
* OK [UIDNEXT 5] Predicted next UID
B OK [READ-WRITE] Select completed (0.001 + 0.000 secs).
C FETCH 3 body[]
* 3 FETCH (BODY[] {445}
Return-path: <user@client.thm>
Envelope-to: strategos@server.thm
Delivery-date: Thu, 27 Jun 2024 16:19:35 +0000
Received: from [10.11.81.126] (helo=client.thm)
        by example.thm with smtp (Exim 4.95)
        (envelope-from <user@client.thm>)
        id 1sMrpq-0001Ah-UT
        for strategos@server.thm;
        Thu, 27 Jun 2024 16:19:35 +0000
From: user@client.thm
To: strategos@server.thm
Subject: Telnet email

Hello. I am using telnet to send you an email!
)
C OK Fetch completed (0.001 + 0.000 secs).
D LOGOUT
* BYE Logging out
D OK Logout completed (0.001 + 0.000 secs).
Connection closed by foreign host.
```

```
Hello. I am using telnet to
send you an email!
```

**5. Logout**

```
D LOGOUT
```

Server:
* BYE Logging out

## IMAP Command Tags (A, B, C…)

A tag is an identifier placed before each command.

Example:

- A LOGIN strategos password
- B SELECT inbox
- C FETCH 3 body[]
- D LOGOUT

They help track which response belongs to which command.

IMAP allows multiple commands to be processed simultaneously.
Because of this, the server responses might arrive in a different order.
Tags ensure the client knows which response corresponds to which command.

Example concept:

```
A LOGIN strategos password
B SELECT inbox
```

Server response:

```
A OK Logged in
B OK Select completed
```

The tag A matches LOGIN, and tag B matches SELECT.

**Common Network Protocols:**

| Protocol | Purpose | Transport Protocol | Default Port |
|---|---|---|---|
| TELNET | Remote command-line access | TCP | 23 |
| DNS | Resolve domain names to IP address | UDP / TCP | 53 |
| HTTP | Transfer web pages | TCP | 80 |
| HTTPS | Secure web communication | TCP | 443 |
| FTP | File transfer between Client and server | TCP | 21 |
| SMTP | Send emails | TCP | 25 |
| POP3 | Retrieve / Download emails | TCP | 110 |
| IMAP | Synchronize emails | TCP | 143 |